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內視鏡超音波影片中標誌性血管的分類與分割

Detection and Segmentation of Landmark Blood Vessels in EUS Video

李天敏

Timothy Buxton Lee

指導教授：陳宏銘 博士

Advisor: Homer H. Chen, Ph.D.

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Detection and Segmentation of Landmark Blood Vessels in EUS Video

本論文係 李天敏 (姓名) R12942161（學號）在國立臺灣大學電信工程學研究所完成之碩士學位論文，於民國 115 年 1 月 26 日承下列考試委員審查通過及口試及格，特此證明。

The undersigned, appointed by the Graduate Institute of Communication Engineering on 26th (month) January (year) 2026 have examined a Master's Thesis entitled above presented by Timothy Lee (name) R12942161 (student ID) candidate and hereby certify that it is worthy of acceptance.

口試委員 Oral examination committee:

陳宏銘 （指導教授 Advisor）　　嚴先晨

李佩君　　林澤　　施光祖

所長 Director:

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中文摘要

在過去的一個世紀中，超音波成像已確立其作為關鍵診斷影像技術的地位。在近四十年內，內視鏡超音波（Endoscopic Ultrasound, EUS）進一步擴展了超音波成像的能力與應用彈性。然而，在執行 EUS 時，一項主要挑戰在於於探頭移動的情況下，如何快速且容易地定位可靠的解剖學標誌。本研究將深度學習應用於 EUS 探頭偏離標準化位置時，在 EUS 影片中偵測標誌性血管的問題。我們提出的方法能提升單一輸出分割模型的中心線準確度，使其表現可媲美多任務分割與中心線模型，並具有以下貢獻：第一，我們提出 C-Dice，作為 centerline Dice 損失函數的延伸；第二，我們建立一個新的 EUS 影片資料集，包含標誌性血管的分類與分割標註；第三，我們探討在 EUS 分割訓練中，將 EUS 分類作為預適應步驟所帶來的效益。結果代表著在創建用於 EUS 的動態術中位置估計系統方面邁出了有希望的一步。

關鍵字：深度學習、影像處理、醫學影像處理、超音波成像、內視鏡超音波

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ABSTRACT

Over the past century, ultrasound imaging has cemented itself as a key diagnostic imaging modality. In the past 40 years, endoscopic ultrasound (EUS) has further expanded the power and versatility of ultrasound imaging. A key challenge in performing EUS is quickly and easily locating reliable anatomical landmarks, especially when the probe is moving. This work applies deep learning to the challenge of finding landmark blood vessels in EUS videos when the EUS probe is away from standardized positions. Our proposed method improves the centerline accuracy of a single-output segmentation model to those comparable to a multi-task segmentation and centerline model, making the following contributions: First, we propose C-Dice, an expansion on the centerline Dice loss function. Second, we introduce a novel dataset of EUS videos with classification and segmentation labels for landmark blood vessels. Third, we investigate the benefits of EUS classification as a pre-adaptation step in EUS segmentation training. These results represent a promising step forward towards creating a dynamic, intra-operative position estimation system for EUS.

Keywords: Deep Learning, Image Processing, Medical Image Processing, Ultrasound Imaging, Endoscopic Ultrasound

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Chapter 1 Introduction

During an EUS procedure, the operator — a specially-trained doctor, nurse, or medical technician — must rely on tactile feedback from the endoscope and visual feedback from the EUS video to intuit the ultrasound probe's position and orientation in 3D space. This is aided by the recognition of anatomical landmarks, structures which are generally static within the body — often blood vessels [4]. Because these landmarks are often visually similar to each other and surrounding anatomy, and because operators must become familiar with the ultrasound imaging of diverse anatomy while simultaneously honing their manual skill with an endoscope, trainees can perform hundreds of supervised procedures before achieving basic competency [5]. The current American Society of Gastrointestinal Endoscopy guidelines recommend completing at least 225 supervised procedures before attempting accreditation as an EUS operator [6].

Fortunately, advances in deep learning image processing have the potential to reduce the cognitive load of EUS procedures and speed up skill acquisition in EUS training. Modern deep learning image classification and segmentation models can be successfully applied to multiple EUS tasks [7], including the detection of landmark and target anatomy [8], [9], [10]. Initial pilot studies on the use of these models suggest that such models can improve skill acquisition in EUS examinations, reducing the number of supervised procedures necessary to reach performance benchmarks [11], [12]. Nevertheless, automatic EUS anatomic segmentation remains a difficult task, due to the same challenges of human interpretation of EUS imagery. Relevant anatomy is easy to recognize from standardized, preset positions (sometimes called "stations"). Existing work on semantic segmentation only trains models to distinguish landmark anatomy at common stations, where visible organs are highly predictable, and does not provide direct

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support in navigating between stations.

We attempt to solve this problem by providing continuous segmentation of landmark blood vessels, even as the probe navigates between EUS stations. We focus on blood vessels because operators can follow landmark vessels to reach different EUS stations [4]. Landmark blood vessels are also chosen because their positions in 3D space are predictable, opening the possibility of continuous, automated probe localization.

In order to support future position tracking and localization work, we investigate training methods that prioritize centerline accuracy. Centerline accuracy is important because tissue outlines are often fuzzy or indistinct in ultrasound video, especially blood vessels and other tubular structures. For example, if a blood vessel curves away perpendicular to the image plane, the blood vessel can appear to blend into the surrounding tissue, with no obvious dividing line between liquid blood and solid tissue. Moreover, because blood vessels are flexible, the vessel wall can be deformed by movement or even blood pressure, while the vessel centerline is remains more defined. Centerline contour matching is thus an easier approach to delineating blood vessels in abdominal ultrasound and a common method for 3D vessel registration. If segmentation maintains a consistent centerline, it is also easier for an observer to maintain focus on a highlighted blood vessel.

We propose a novel system for EUS semantic segmentation focused on the accurate segmentation of landmark blood vessels from multiple angles. This is intended to continuously guide an EUS operator towards common stations, rather than just confirm when they have successfully reached a station. In this, we make the following contributions: First, we demonstrate the feasibility of classifying landmark blood vessels while the EUS probe is in motion, away from stations. Second, we investigate the advantage of using EUS anatomical classification as a segmentation backbone versus a

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popular backbone. Third, we present C-Dice (short for centrality Dice), a novel loss function based on the centerline Dice (clDice) metric and loss function [13], a topology-preserving loss function for segmenting tubular objects. C-Dice achieves better volumetric and centerline accuracy than ordinary clDice loss, and can even produce better-aligned centerline predictions than a multi-task segmentation and centerline estimation model.

We demonstrate these contributions on a newly-developed dataset, produced in collaboration with National Taiwan University Hospital, consisting of real recordings of EUS examinations with classification and segmentation labels for the portal vein confluence, a set of landmark blood vessels. For classification, we trained a ResNet-50 model [14], which was incorporated into the segmentation model as a backbone. For segmentation, a UNet3+ decoder [15] was trained with our modified C-Dice loss function and results were compared against both existing clDice methods and traditional segmentation loss. The performance of C-Dice was validated on two external segmentation tasks: blood vessels in retinal fundus photography and the left chambers of the heart in echocardiograms.

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Chapter 2 Related Work



2.1 Deep Learning in EUS

The application of deep learning to EUS imaging shows general promise, with many groups reporting success in diagnostic tasks such as tumor localization, cyst characterization, and biopsy guidance [7]. With sufficient training data, modern deep learning models can reach comparable classification and segmentation accuracy as accredited doctors [7]. Studies focused on anatomical classification and segmentation, on the other hand, remain sparse.

Pilot studies [8] and [9] summarize conference presentations of object detection models, one unspecified model and one YOLOv11 model. Both successfully detected important anatomy in EUS video, achieving F_1 scores of 92.0% and 92.5%. In a more detailed paper, classification only was attempted on a set of five anatomical structures, adding observer voice commands to improve accuracy; however, the chosen anatomical structures proved more difficult to distinguish, only achieving an F_1 score of 74.0% [10].

2.2 Anatomic Segmentation of EUS Video

For semantic segmentation, [11] and [12] presented two iterations of a multi-stage classification and segmentation design, which (1) first used a ResNet-50 classification model to validate image frames containing useful information, then (2a) used a separate ResNet-50 classification model to estimate whether the ultrasound probe is at predetermined stations and (2b) a UNet++ model [16] to segment target anatomy. This achieved a Dice score of 77% for the pancreas and 81.3% for target blood vessels in [11] and 77% for the bile ducts in [12]. However, the segmentation model displayed no adaptations particular to semantic segmentation in EUS video, as these models were

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prepared using only standardized pre-trained weights and cross-entropy loss. Of particular note, in [11] the segmentation model was only trained to recognize landmark blood vessels at two out of six stations, with blood vessel segmentation performance outside those stations unreported. In [12], the bile ducts were only segmented when the doppler imaging false color overlay was enabled — essentially, tubular structures were only segmented when they were clearly identifiable. Nevertheless, the overarching design showed promise, as both applications also found that their segmentation results had similar performance to the inter-operator agreement of a group of experts, and crossover studies in both cases (with 8 and 12 trainees, respectively) saw improvement in both the speed at which trainees recognized preset positions and annotation accuracy of target anatomy in EUS video. These initial studies support the potential for deep learning image processing techniques to promote skill acquisition in EUS training, while leaving room for further improvement on the models.

2.3 clDice Metric

Given that several of the anatomical landmarks for upper body EUS are blood vessels, one potential avenue to improve segmentation stability and accuracy is the introduction of deep learning techniques used to segment blood vessels in other applications. Centerline Dice (clDice) [13] is a metric introduced by Shit et al. to help evaluate segmentation of tubular structures, such as blood vessels. While the conventional Dice score measures the F_1 score of the full segmentation volumes, clDice measures the proportion of the predicted segmentation's centerline lies within the ground truth volume and vice-versa, treating the two values as the precision and sensitivity for F_1 score calculation.

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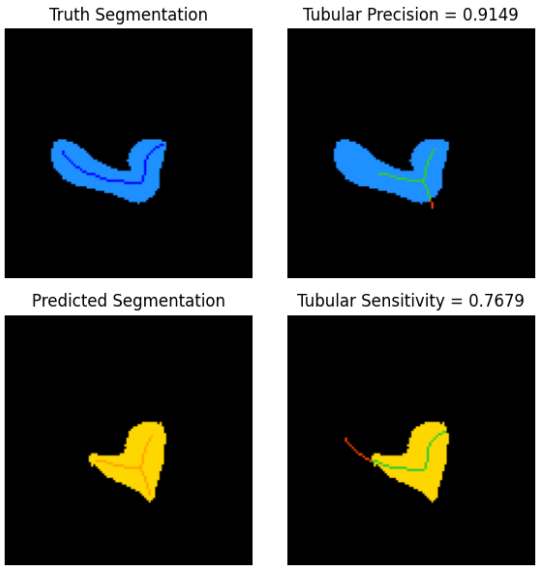


Fig. 1.1. Tubular precision and tubular sensitivity.

clDice attempts to address the issue of segmentation models achieving high pixel-wise or volumetric accuracy, but poor topological accuracy. As a loss function, this is used in conjunction with ordinary Dice loss, as shown below:

$$L_{Dice} = 1 - \frac{2 \cdot |X \cap Y|}{|X| + |Y|}$$

Eq. 1.1 Dice loss formula.

$$Tprec = \frac{|X_C \cap Y|}{|X_C|}$$

Eq. 2.1. Tubular precision formula, the first component of clDice. X_C denotes the predicted centerline and Y denotes the ground truth volume.

$$Tsens = \frac{|Y_C \cap X|}{|Y_C|}$$

Eq. 2.2 Tubular sensitivity formula, the second component of clDice. X denotes the

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predicted volume and Y_C denotes the ground truth centerline.

$$clDice = \frac{2 \cdot (Tprec \cdot Tsens)}{Tprec + Tsens}, L_{clDice} = 1 - clDice$$

Eq. 2.3. clDice metric and loss formulae.

$$L_C = (1 - a) \cdot L_{Dice} + a \cdot L_{clDice};\ 0 < a < 1$$

Eq. 2.4 Full loss function use in [13]

2.4 clDice Loss and its Successors

With a deep learning-compatible centerline extraction method, clDice loss can improve topological accuracy, and in some instances improve volumetric accuracy as well. Of note, the initial clDice paper [13] did not use a true centerline extraction method, but instead approximated the centerline using minpooling and maxpooling to affect erosion and dilation; eroded points which are not restored by dilation are designated as the centerline. The authors dubbed this technique “soft-skeletonization” and its corresponding loss function soft-clDice loss. The main limitation of soft-skeletonization is that it creates gaps in the estimated centerline, as shown in Fig 2.1. Nevertheless, in all original test cases, soft-clDice loss increased accuracy on all measures of centerline alignment and topological accuracy [13]. A smaller limitation of the original implementation of soft-clDice loss is that was it only demonstrated on very thin structures, such as blood vessels in retinal photographs or roads in satellite images. Later work would demonstrate that soft-skeletonization, and thus soft-clDice as well, is less consistent when applied to larger structures.

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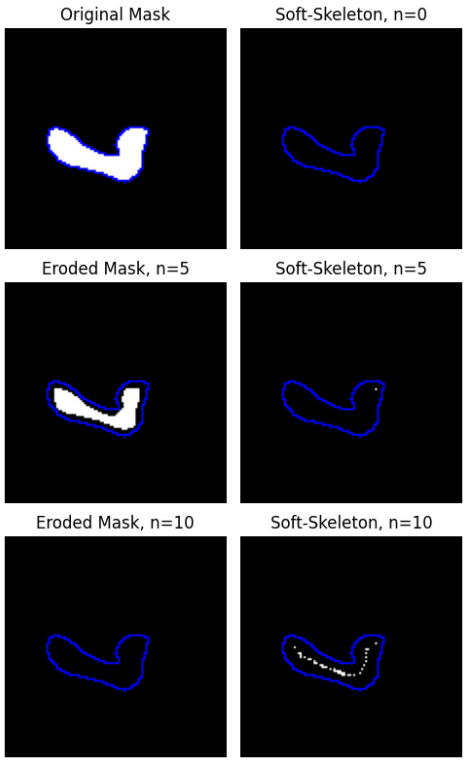


Fig. 2.1. A demonstration of soft-skeletonization

Since this publication, multiple attempts have been made to improve the implementation of clDice loss and adapt it to more image modalities. In [17], the centerline approximation method was replaced with an erosion method based on *simple points*. Simple points are points of a multidimensional shape which can be deleted without changing the shape's topological connectivity [18]. Normally, simple point erosion is calculated on a pixel-by-pixel basis, but such calculations create too many gradient steps for practical deep learning. The authors of [17] demonstrate that parallelizing simple point erosion over staggered subsamples of an image produces more complete skeletons while still using a manageable number of gradient steps for deep learning.

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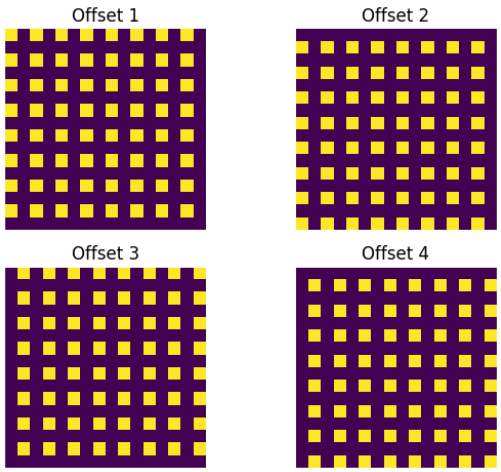


Fig. 3.1. The four sample grids used by [17]'s proposed parallelized erosion method

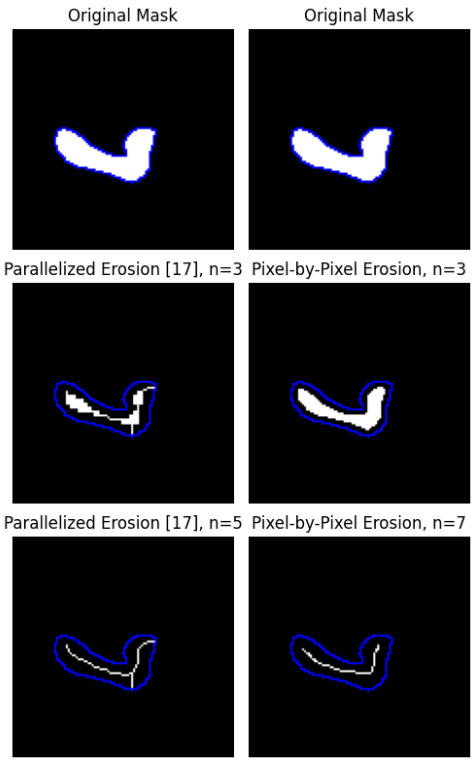


Fig. 3.2. Visual comparison of parallel vs pixel-by-pixel simple point erosion

So far, all iterations on clDice loss use either soft-skeletonization or parallelized simple point erosion. Despite its higher centerline accuracy, [17] remains less popular than soft-clDice loss due to higher computational demands.

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For instance, [19] and [20] were built directly upon the soft-skeletonization algorithm and achieved nominal improvements. In [19], the gaps in soft-skeletonization's outputs are filled via dilation with a diamond-shaped 5x5 kernel, then only use this region's tubular sensitivity, not clDice, in the final loss function. This achieved higher Dice and clDice scores than both Dice loss alone and soft-clDice in four out of five datasets, and was successfully trained on an additional 3D multi-organ CT segmentation dataset whereas soft-clDice ran out of memory. However, this method only was less consistent at improving topological connectivity. The authors also compared the results against the rigorous method in [17] on three datasets, which achieved comparable Dice and clDice scores, but only successfully reduced Betti-0 error in one out of three datasets.

In [20] the authors use the outputs of soft-skeletonization to sample the pixel-wise cross-entropy loss between prediction and ground truth, combining this with conventional Dice loss alone to train their model. Across five retinal blood vessel datasets, this achieved the highest Dice score versus baseline Dice and soft-clDice loss. On the other hand, this centerline cross-entropy loss generally had markedly lower clDice score gains versus soft-clDice.

Finally, [21] sought to utilize the more robust algorithm in [17] to create a weighted clDice based on multiple calculations of boundary distance and radius. This improved not only topological accuracy but also reduced the average distance between prediction and ground truth. Unfortunately, there was a notable drawback in processing requirements. The authors noted a batch size of only 2. Although the designs did achieve the best scores across all metrics in the evaluated 2D dataset and the overall best metrics in two out of three 3D blood vessel segmentation tasks [20], the Dice score was only slightly lower than the method in [17] for the final task. Nevertheless, the increased training overhead required for this approach means that it cannot be the most reliable

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choice in all applications.



2.5 clDice and Multi-Task Learning

Ordinarily, segmentation and centerline estimation are treated as separate deep learning tasks performed with multi-task models using cascaded architectures [22] or parallel outputs. Although segmentation and centerline estimation are trained simultaneously, they are separate model outputs. Since its development, clDice has also been used to refine such models to general success. Liu et al. proposed a multi-decoder UNet architecture with additive feature fusion blocks to improve consistency between segmentation and centerline outputs [23], trained with Dice and clDice loss, and achieved lower average centerline distance. Another paper demonstrated that cascaded multi-task networks can also potentially outperform single-output models when trained with a blend of Dice and clDice loss, albeit less consistently [24].

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Chapter 3 Proposed Method

Accurately extracting blood vessel centerlines is key to the further development of position estimation in EUS video. An accurate centerline prediction implicitly yields accurate information on the blood vessel's spatial orientation. We propose the introduction of a centerline loss function to EUS blood vessel segmentation to improve centerline connectivity and alignment. The completeness of the centerline takes priority over volumetric accuracy, because the centerline provides more information on the orientation of a blood vessel than its area. Centerline completeness also requires accurate centerline alignment to provide accurate information on blood vessel orientation. Additionally, training an EUS segmentation model with centerline loss should indirectly promote smoother segmentation in areas with low detection confidence by penalizing disjointed or noisy predictions.

In this section we present a new loss function, C-Dice, which achieves greater centerline accuracy than previous methods, particularly on thicker structures than clDice was originally designed for. The proposed method retains the intermediate steps of skeletonization. Importantly, these intermediate steps provide information of pixels' distance to the centerline, contributing shape alignment information to the final training loss calculation.

3.1 Difficulties of Centerline Loss in EUS

The main shortcoming in adapting clDice loss to larger structures, such as blood vessels in EUS video, is that tubular precision (Eq. 2.1) and tubular sensitivity (Eq. 2.2) become increasingly permissive as the segment radius increases. In short, the existing method is both less accurate and less robust for larger blood vessels. This is because, as the radius increases, the skeleton occupies a smaller proportion of a segment's pixels,

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allowing more variety of skeleton shapes to fit inside the target segmentation and vice-versa. It is possible for conventional Dice and clDice to evaluate equal loss on two segmentations with wildly different centerline size and shape, as shown in Fig. 4.1. In these instances, variations with wildly different centerline distance error will still equate in the same magnitude of clDice loss. This dilutes the effect of other loss functions used in the final training loss.



Fig. 4.1. Dice and clDice alone do not directly relate to centerline alignment

3.2 C-Dice Loss Function

Maximizing clDice's improvement to EUS segmentation will require a new loss function that accounts for centerline alignment within the target area. To accomplish this, we measure each segment pixel's distance to the centerline, thereby creating a contour map, weighted from 0.0 to 1.0. Pixels on the centerline are assigned a value of 1.0, with value decreasing with distance; the rate of decrease is proportional to the radius of the segment. Background pixels are set to a value of 0.0. A new Dice-based metric is calculated, based on the precision and sensitivity of these contour maps (see Eq. 3.1-3.3). We call the new resulting loss function C-Dice, short for centrality Dice, as the contour map closely correlates with pixel centrality to the centerline.

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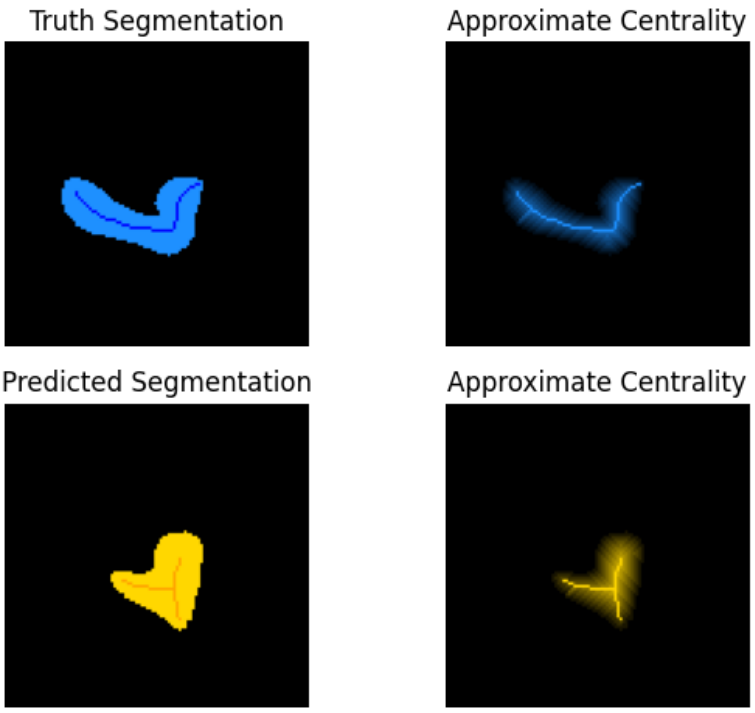


Fig. 5.1. The proposed centerline contour map

In physical terms, C-Dice measures the mass distribution of an object, whereas clDice measures only the center of gravity. On small objects, these are roughly equal, but on larger objects, variations in shape are more clearly measured in the mass distribution. C-Dice has greater sensitivity to changes in radius and asymmetric structural complexity than clDice, providing more meaningful information about centerline accuracy.



Fig. 6.1. C-Dice does measure centerline alignment

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$$CPrec = \frac{|C_X \cap Y|}{|C_X|}$$

Eq. 3.1 Centrality precision formula

$$CSens = \frac{|C_Y \cap X|}{|C_Y|}$$

Eq. 3.2 Centrality sensitivity formula

$$L_{C-Dice} = 1 - \frac{2 \cdot (Cprec \cdot Csens)}{Cprec + Csens}$$

Eq. 3.3 Centrality Dice loss formula

$$L_C = (1 - \alpha) \cdot L_{Dice} + \alpha \cdot L_{C-Dice};\ 0 \leq \alpha \leq 1$$

Eq. 3.4 Dice and C-Dice combined loss formula with a balancing coefficient α

To ensure differentiability for gradient backpropagation and maximize computational efficiency, we further propose to use the order of pixel erosion during skeletonization as a proxy metric for centerline distance. In erosion-based skeletonization, each full pass over an image will erode a single layer of non-centerline border pixels. Ordinarily, intermediate passes from before complete skeletonization are discarded, but storing these intermediate images can enable the creation of the centrality contour map with a minimum of additional steps, as follows:

Typical Procedure:	C-Dice Procedure:
1. Given an input image I, a differentiable skeletonization method S 2. $I' = S(I)$ 3. $I = I'$	1. Given an input image I, a differentiable skeletonization method S, and a blank output image O 2. $I' = S(I)$

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4. Repeat step 2 and 3 until $I = S(I)$ or maximum iterations reached 5. Return I as output	3. $I = I'$ 4. $O = O + I$ 5. Repeat step 2-4 until $I = S(I)$ or maximum iterations reached 6. $O = O/\max(O)$ 7. Return O as output

Table 1.1. Comparison of the skeletonization procedure used by clDice versus the erosion contour map created by C-Dice

Any differentiable skeletonization or erosion method may be used to implement C-Dice loss. For our final implementation, we apply the simple point erosion method from [17] for its greater centerline accuracy. To streamline the combining of thinned masks and accelerate overall training time, an early-stopping mechanism was put in place to prevent further simple point calculations when no simple points remain, unlike previous implementations [13], [17]. This is tracked for each individual image in a batch, further simplifying backpropagation. An upper bound on the number of iterations is set proportional to the maximum number of iterations needed to fully skeletonize all images in the training set, as used in previous work [13], [17].

For the final calculation of training loss, cross-entropy loss (CE), standard Dice loss, and C-Dice were combined according to two balancing coefficients, α and β, which defines the ratio of Dice to C-Dice and CE loss to Dice-based loss.

$$L = \beta \cdot L_{CE} + (1-\beta) \cdot ((1-\alpha) \cdot L_{Dice} + \alpha \cdot L_{C-Dice});\ 0 \le \alpha \le 1, 0 \le \beta \le 1$$

Eq. 4.1 Total loss function used with balancing coefficients α and β

3.3 Model Pre-Training

It is well-established that pre-training deep learning models on related tasks can improve model performance and reduce training data requirements for the final model.

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However, there are currently no pre-trained models specific to ultrasound imaging. We propose that task-specific classification pre-training can bridge the gap between high quality general-purpose encoders and ultrasound segmentation tasks. Classification labeling of video data is substantially easier than segmentation labeling, and can be used as a powerful reference to accelerate and improve subsequent segmentation labeling.

In our application, the portal vein confluence (PVC) is a widely-used EUS landmark composed of three connected blood vessels. The component vessels each have look-alike structures in the same region of interest. We propose to first train a model to detect the individual blood vessels of the PVC from the moment they enter frame and, perhaps more importantly, reject images containing only look-alike structures. A model with this encoder should more readily distinguish target from non-target anatomy, reducing the rate of false detections.

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Chapter 4 Experiment



4.1 Data Collection

Data collection was performed in collaboration with the Division of Gastroenterology and the Department of Medical Imaging, National Taiwan University Hospital (NTUH), from September 2024 through April 2025. NTUH doctors approached patients about to undergo ordinary EUS examinations of the liver, pancreas, or bile ducts to attain informed consent for the release of EUS recordings for research purposes. Additional patients were added to the study pool via records search, in accordance with contemporary rules and regulations. All data were fully anonymized, retaining only the date of procedure and the NTUH patient ID number.

In total, 49 EUS recordings were collected from 37 patients. Recordings were selected from patients without major anatomic abnormalities or pathologies. During and after video collection, doctors at the NTUH Division of Gastroenterology highlighted the portal vein, splenic vein, and superior mesenteric vein when the probe is near a common station. Due to logistical limitations in data collection and annotation, two graduate students at NTU Graduate Institute of Communications were taught to recognize these blood vessels in EUS video, and prepared machine learning-compatible labeling of the target blood vessels. During this phase, labeling scope was expanded to include sequences where target blood vessels enter or exit the image frame. EUS videos in which blood vessels were not readily distinguishable from surrounding tissue, even after contrast adjustment, were excluded from the study pool. Before final experiments, label accuracy was validated by NTUH doctors. The final dataset included 32 videos.

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There were two stages of EUS labeling, corresponding to later two stages of deep learning. In order to support the pretraining of the model backbone, images were given classification labels for “PV Only,” “SMV Only,” and “PV-SV-SMV,” corresponding to images where the only visible target vessels were the portal vein, superior mesenteric vein, or the portal vein confluence, respectively. The portal vein confluence was distinguished by whether curvature or narrowing was visible at the blood vessel junction. For the final model, polygon segmentation of the target blood vessels was performed using a single segmentation class. Target blood vessels were not distinguished during segmentation because the target vessels form a contiguous structure, wherein the component vessels cannot be reliably separated at all angles.

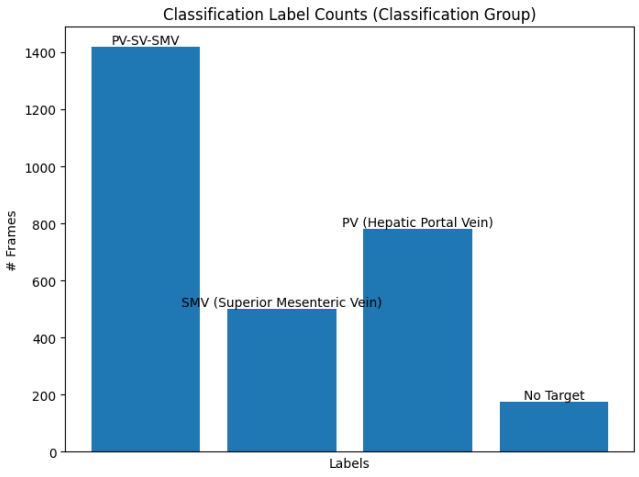


Fig. 7.1. Distribution of frame counts per video for classification labels

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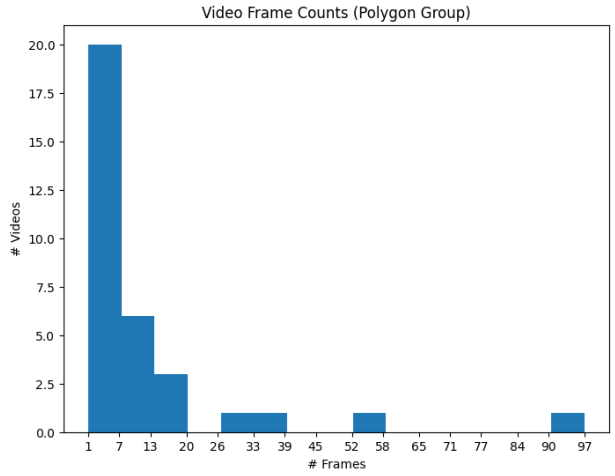


Fig. 7.2. Distribution of frame counts per video for segmentation labels

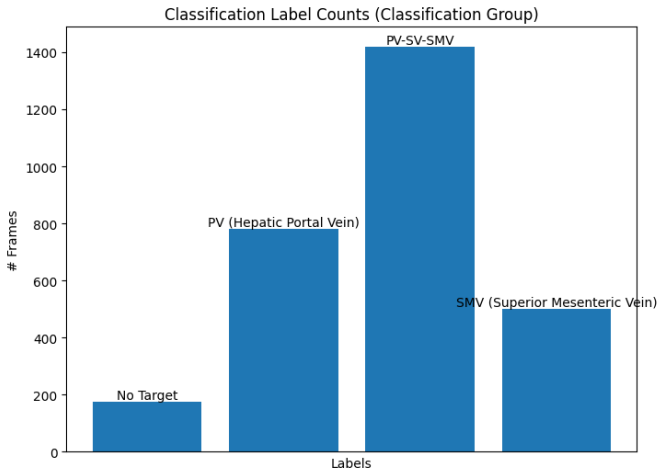
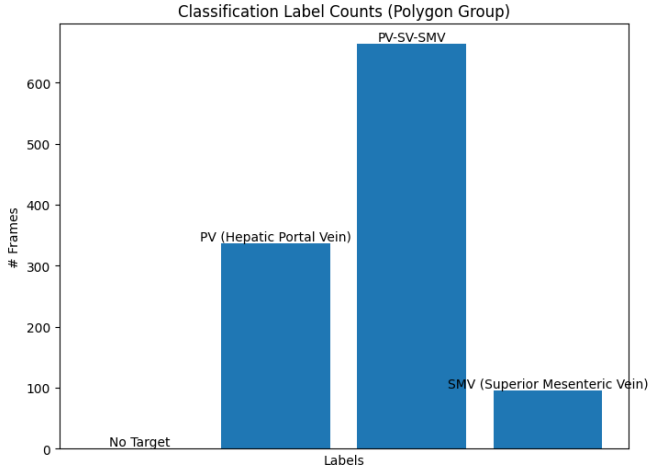


Fig. 7.3. Distribution of classification labels in the classification subset



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Fig. 7.4. Distribution of classification labels in the segmentation subset

Images containing segmentation were then filtered to produce sequences of four consecutive frames, with permissible overlap of two frames; at least two frames must contain unique polygon segmentations. This reduces the number of sequences where the EUS probe is stationary, improving image diversity. The segmentation sequences were then incorporated into the backbone classification dataset, producing 1,132 segmentation images from 410 segmentation sequences and 2,873 classification labels. Training, validation, and test splits were divided by an 8:1:1 ratio. Before training, polygon segmentations were converted into semantic segmentation masks.

4.2 Model Training

4.2.1 Data Augmentation

Various image transformations were used to augment the training data. Images were preprocessed with quantile-based contrast stretching, normalizing image intensity across different ultrasound contrast settings. During training, images were randomly cropped or shrunk by up to +/- 15% and randomly rotated by up to +/- 15°. These values are smaller than is often used in other applications to ensure that target blood vessels always remain within the augmented image, and these transformations are re-centered to avoid distorting the image's curvilinear layout. Images were then randomly flipped along the horizontal axis with p = 0.5, and finally resized to 224x224 pixels.

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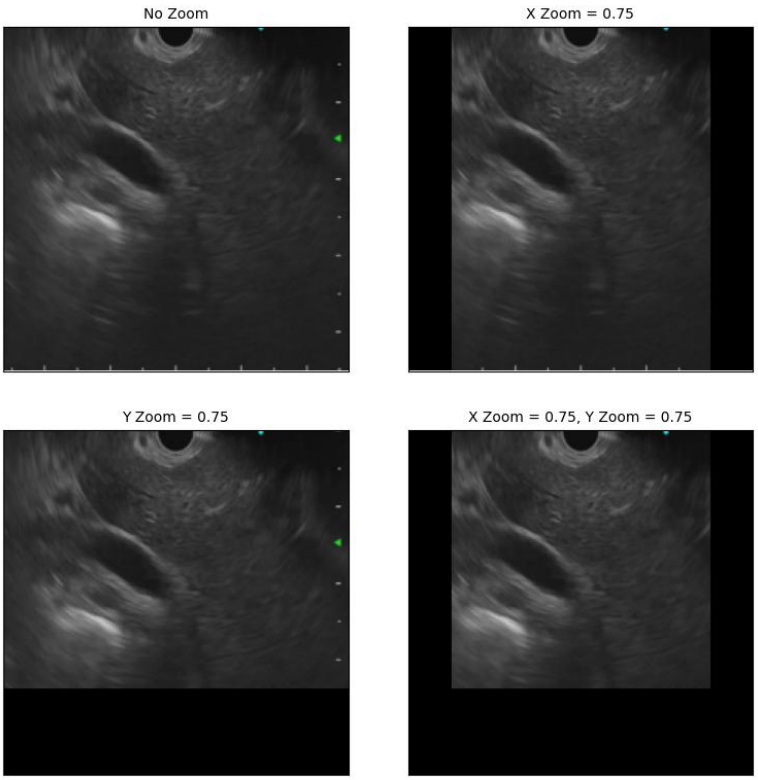


Fig. 8.1. Ultrasound image crop methods



Fig. 8.2. Ultrasound image rotation methods

4.2.2 Classification Backbone

The EUS classification backbone was trained for 90 epochs and batch size of 32 with stochastic gradient descent and weighted cross-entropy loss. The optimizer used a base learning rate of 0.01 on a cosine-annealing schedule with momentum of 0.9 and weight decay of 0.0001. The Cross-entropy weighting was proportional to the class prevalence. Models were initialized with the Torchvision V1 pre-trained ImageNet-1K classification

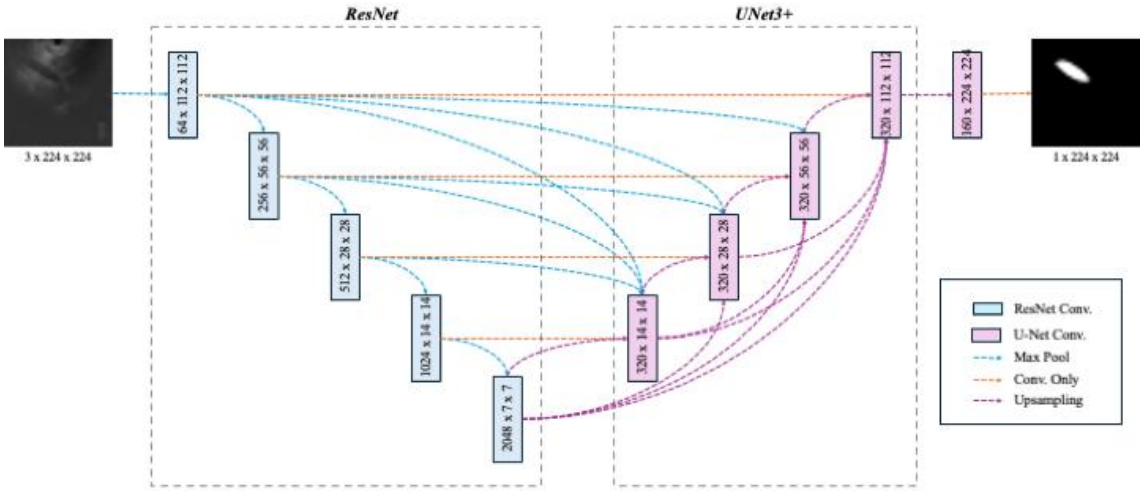
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weights [25], [26].

Classification training was used to estimate the ideal quantile thresholds for contrast stretching, testing minimum quantile values of 0.025 and 0.050 and maximum quantile values of 0.9900, 0.9950, and 0.9975. The quantile range of [0.025, 0.9975] achieved the best accuracy and F1 score, and the preprocessed images from this test were preserved for segmentation training.

4.2.3 Segmentation Decoder

For segmentation, we use a UNet3+ decoder [15] adapted for the ResNet-50 backbone. As ResNet uses one additional resolution downsampling step than most UNet methods, we append an additional UNet upsampling module before final outputs. We use the UNet3+ architecture instead of the more popular UNet++ architecture because UNet3+ achieved higher Dice scores in preliminary tests. The segmentation decoder was trained for 30 epochs and batch size of 8 with an Adam optimizer and the loss function proposed in Equation 4.1. The learning rate was 0.0001 with a cosine-annealing schedule. To evaluate the impact of EUS classification pre-training, we test the following: (1) a frozen pre-trained backbone, (2) fine-tuning the pre-trained backbone during segmentation training, and (3) fine-tuning the conventional ImageNet-1K backbone.



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Fig. 9.1. ResNet50 encoder UNet3+ decoder architecture

To test C-Dice against existing methods, we compare our loss function against four previous versions of centerline loss and the conventional Dice loss. We test both soft-clDice loss and [17]'s clDice implementation, as well as [19]'s skeleton recall loss and [20]'s centerline cross-entropy loss. Because previous work incorporating centerline loss functions saw improvement with the introduction of cross-entropy loss [20], [21], we also investigate its potential benefits to EUS landmark segmentations.

Label	Method	Formula
A1	Dice	$L = L_{Dice}$
B1	Soft-skeletonization [13]	$L = \frac{L_{Dice}}{2} + \frac{L_{soft-clDice}}{2}$
C1	Simple point erosion [17]	$L = \frac{L_{Dice}}{2} + \frac{L_{clDice}}{2}$
D1	Soft-skeleton recall [19]	$L = \frac{L_{Dice}}{2} - \frac{Tsens}{2}$
E1	Soft-skeleton cross-entropy [20]	$L = \frac{L_{Dice}}{2} + \frac{CE_{Tprec} + CE_{Tsens}}{2}$
F1	Simple point erosion *(ours)*	$L = \frac{L_{Dice}}{2} + \frac{L_{C-Dice}}{2}$
A2	Cross-Entropy + Dice	$L = \frac{L_{CE}}{2} + \frac{L_{Dice}}{2}$
B2	Cross-Entropy + Soft-skeletonization [13]	$L = \frac{L_{CE}}{2} + \frac{L_{Dice}}{4} + \frac{L_{soft-clDice}}{4}$

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C2	Cross-Entropy + Simple point erosion [17]	$L = \frac{L_{CE}}{2} + \frac{L_{Dice}}{4} + \frac{L_{clDice}}{4}$
D2	Cross-Entropy + Soft-skeleton recall [19]	$L = \frac{L_{CE}}{2} + \frac{L_{Dice}}{4} - \frac{Tsens}{4}$
E2	Cross-Entropy + Soft-skeleton cross-entropy [20]	$L = \frac{L_{CE}}{2} + \frac{L_{Dice}}{4} + \frac{CE_{Tprec}+CE_{Tsens}}{4}$
F2	Cross-Entropy + Simple point erosion *(ours)*	$L = \frac{L_{CE}}{2} + \frac{L_{Dice}}{4} + \frac{L_{C-Dice}}{4}$

Table 2.1. Loss functions used in experiment.

4.3 External Models

For comparison, we implement the clDice-trained multi-task models described in [23], [24] and [27], using the loss functions from their original papers. [23] describes a multi-decoder UNet with feature crossover between decoders called DTUNet. [24] uses CasNet [22] retrained with soft-clDice [13]. [27] describes a similar model to [23]'s DTUNet, but does not use any form of clDice loss. We also include the multi-decoder UNet used as a baseline in [23], trained on that paper's loss function.

For all multi-task models, the encoder is substituted for the pre-trained ResNet-50 backbone from our pre-training experiment.

4.4 Evaluation Metrics

The performance of the classification model was evaluated via overall accuracy and the mean F1 score for each label.

To evaluate the performance of C-Dice, we examine the volumetric accuracy using the Dice score. Centerline accuracy was evaluated using the clDice, C-Dice, Chamfer distance (CD), and sliced Wasserstein distance (SWD). Topological connectivity was evaluated by Betti-0 (β_0) and Betti-1 (β_1) errors. The evaluation used rigorous point-

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by-point skeletonization on both the prediction and ground truth. CD takes the sum of the nearest-neighbor distance from both prediction to truth and vice-versa. SWD approximates the Wasserstein distance (also known as earth mover's distance), which describes the optimal transport cost to transform one distribution into another. We use both CD and SWD because CD will more directly reflect the prevalence of inaccurate skeleton branches, while SWD takes a normalized measure of overall alignment. We use the SWD instead of ordinary Wasserstein distance because Wasserstein distance is too computationally expensive for multidimensional data. To ensure convergence towards the true Wasserstein distance, we set a suitably high number of samples, L = 5000.

$$CD = \sum_{x \in S_X} min_{y \in S_Y}(d(x,y)) + \sum_{y \in S_Y} min_{x \in S_X}(d(y,x))$$

Eq. 4.1 Chamfer distance formula for two sets of coordinates, S_X and S_Y.

$$Err_{\beta_0} = \beta_0(Y) - \beta_0(X);\ \beta_0(X) = n.objects\ in\ X$$

Equation 4.2 β_0 error formula for two structures, X and Y.

$$Err_{\beta_1} = \beta_1(Y) - \beta_1(X);\ \beta_1(X) = n.holes\ in\ X$$

Equation 4.3 β_1 error formula for two structures, X and Y.

4.5 External Datasets

For validation purposes, the same model architecture was trained on two external segmentation tasks: retinal blood vessels in color photographs and heart anatomy in echocardiography. (1) Retina blood vessel segmentation is a common test for segmenting tubular structures due to the high structural complexity. Due to the small size of retinal

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photograph datasets, two datasets were combined — CHASE_DB1 [28] and DRIVE [29] — for a total of 48 color images with accompanying blood vessel segmentation masks. (2) Echocardiography is the most well-developed ultrasound segmentation task where the targets are medium-sized tubular structures. The CAMUS dataset [30] contains ultrasound images from 500 patients with accompanying segmentation labels of the left ventricle epicardium, left ventricle endocardium, and left atrium wall. The end-systolic and end-diastolic image groups were used, totaling 2,000 images. For both outside datasets, train, validation, and test splits were divided on an 8:1:1 ratio.

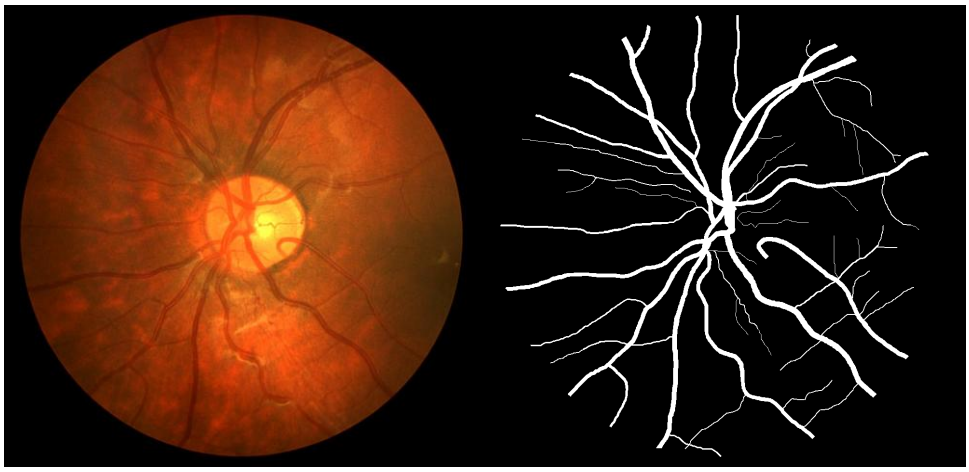


Fig. 10.1. Example image and mask from the retinal images data

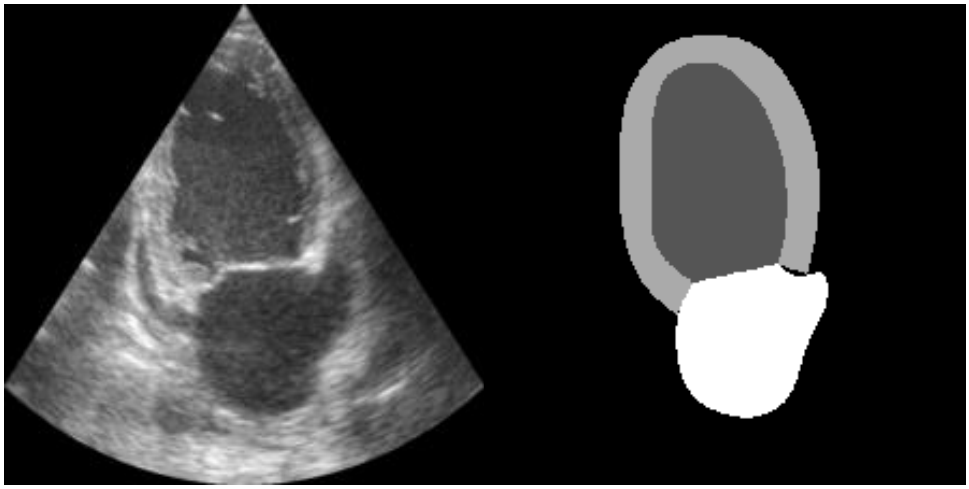
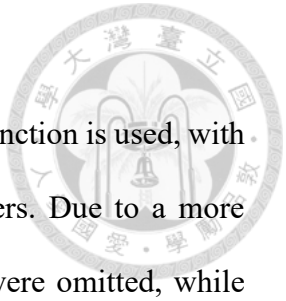


Fig. 10.2. Example image and mask from the echocardiography data

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For external data testing, the same model architecture and loss function is used, with minor modification to the data augmentation and training parameters. Due to a more standardized image field of view, random cropping and shrinking were omitted, while rotation range was increased to up to +/- 180° for retinal photographs and omitted for cardiac ultrasound. Additionally, retinal photographs were resized to 512x512 pixels to preserve segmentation mask quality. Contrast normalization was not used. Due to available GPU sizes, both test cases had a reduced batch size of 4 and an increased base learning rate of 0.0005. For retinal photos, the model was trained for 100 epochs due to the small dataset size.

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Chapter 5 Results

5.1 Contrast Tuning Experiment

Image Contrast		Classification Scores	
Minimum Quantile	Maximum Quantile	Accuracy	F1 Score
0.000	*1.0000*	96.53%	96.51%
0.025	*0.9900*	96.18%	96.17%
0.025	*0.9950*	96.88%	96.84%
0.025	*0.9975*	**97.22%**	**97.21%**
0.050	*0.9900*	96.88%	96.86%
0.050	*0.9950*	96.63%	96.50%
0.050	*0.9975*	96.88%	96.87%

Table 3.1. The performance of the backbone based on contrast settings

The performance difference between different contrast settings was very narrow, but applying minor contrast adjustment was still an improvement in raw accuracy and F1 score over no adjustment. Both the highest raw accuracy and mean F1 score were achieved by removing the top 0.25% and bottom 2.5% of pixel intensities, which was sufficient thresholding to normalize image contrast across videos while retaining the most information.

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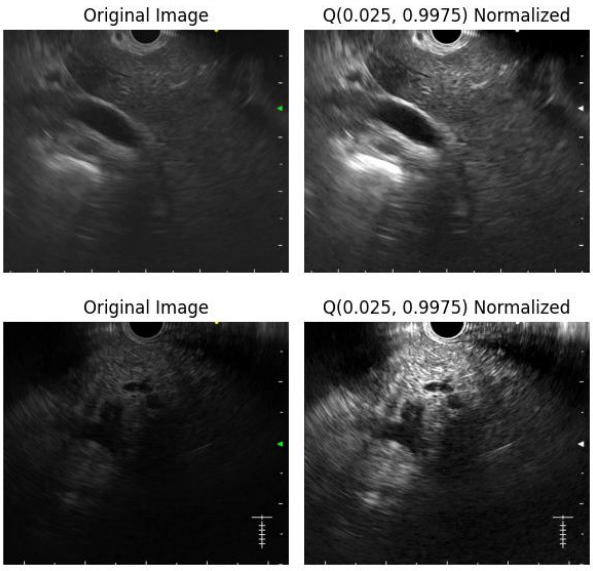


Fig. 11.1. Effects of the chosen quantile-based contrast stretching

5.2 Final Experiment

5.2.1 β=0 Experiment (No Cross-Entropy Loss)

When using structural loss methods only, our loss function achieved the highest Dice, clDice, and C-Dice scores, with 91.17%, 95.92%, and 95.26% respectively. Within each backbone configuration, C-Dice loss produced higher Dice, clDice, and C-Dice scores than each other loss function tested. Examining the outputs, the most obvious benefit of C-Dice loss over other loss functions was a large qualitative smoothness. Other loss functions had significant noise around segmentation boundaries, whereas C-Dice loss produce little-to-no segmentation border noise.

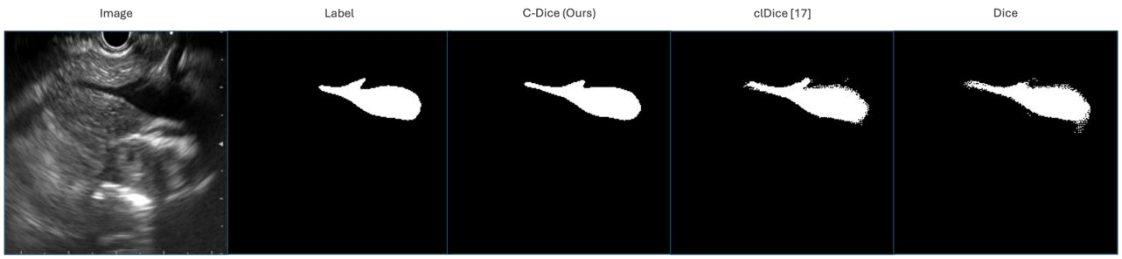


Fig. 11.2. Example segmentation results

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Although the conventional ImageNet-1K backbone weights achieved the highest Dice score, this did not hold true across all loss functions. It also did not achieve equivalent success in measures of centerline accuracy. The fine-tuned pre-adapted backbone achieved higher clDice for both our loss function and soft-clDice, and the same backbone generally achieved lower CD and SWD than the ImageNet-1K backbone. Through this, we demonstrate that pre-training on a highly-related classification task is a potential way to boost segmentation accuracy without expanding the more labor-intensive segmentation labeling process.

5.2.2 β=1 Experiment (Cross-Entropy Loss)

With the addition of cross-entropy loss, the Dice and clDice accuracy of each loss function increased slightly, although the performance of existing centerline loss functions with an ImageNet-1K backbone increased the most. C-Dice loss and our fine-tuned pre-adapted backbone performed second best by a narrow margin in all metrics but C-Dice and Chamfer distance. With no centerline loss, the pre-adapted backbones were more accurate except for by Chamfer distance. The overall pattern is thus: with centerline loss but no centerline alignment loss, backbone pre-adaptation does not out-perform the baseline pre-trained weights. With centerline alignment loss or no centerline loss, backbone pre-adaptation performs better than using baseline pre-trained weights.

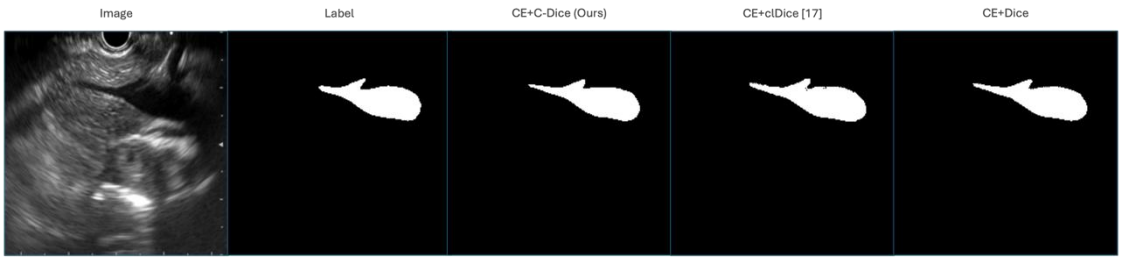


Fig. 11.3. Example segmentation results

Qualitatively, the introduction of cross-entropy loss greatly reduced the

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segmentation border noise from all model outputs, but slightly increased the incidence of false positive segmentations of non-landmark blood vessels. It is possible that, given the size and diversity of the current dataset, the learnt fine-detail features of landmark vessel borders and non-landmark vessels are similar enough that optimizing the pixel-wise predictions for one impacts the other.

The identical performance of soft-clDice and clDice with the ImageNet-1K backbone was investigated and found not to be an error.

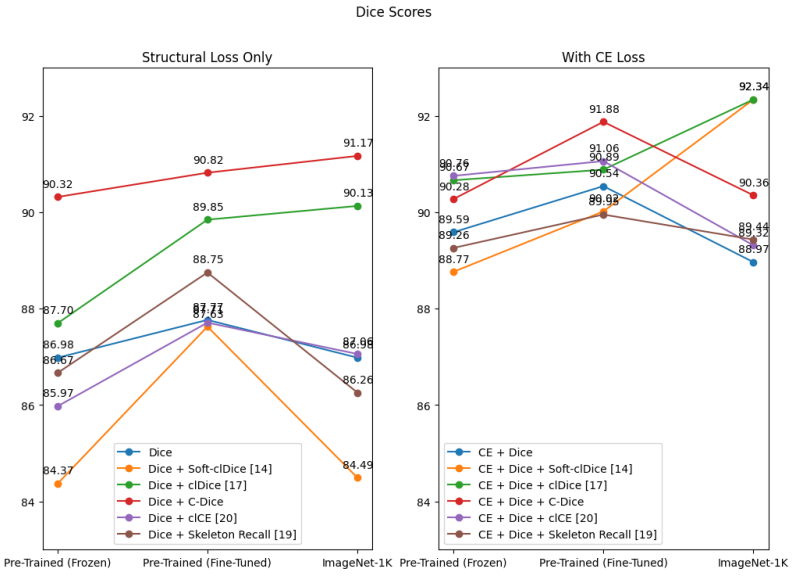


Fig. 12.1. Dice performance of proposed model

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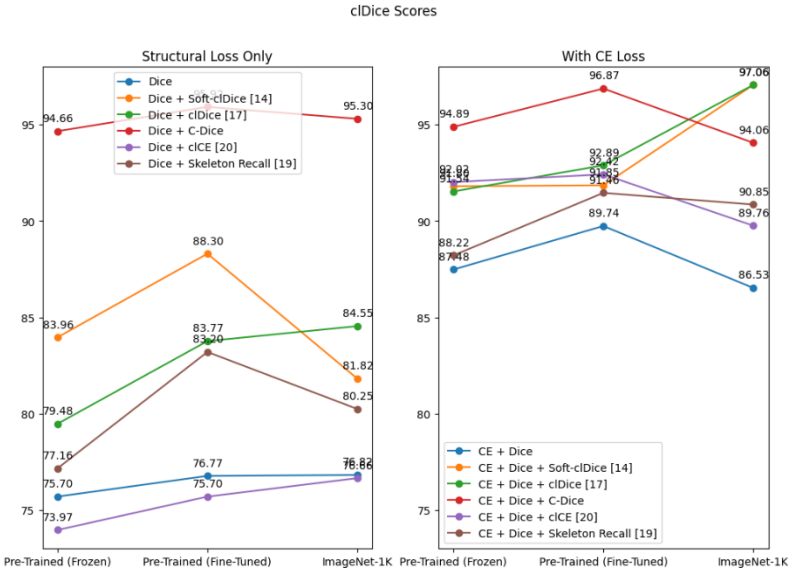


Fig. 12.2. clDice performance of proposed model

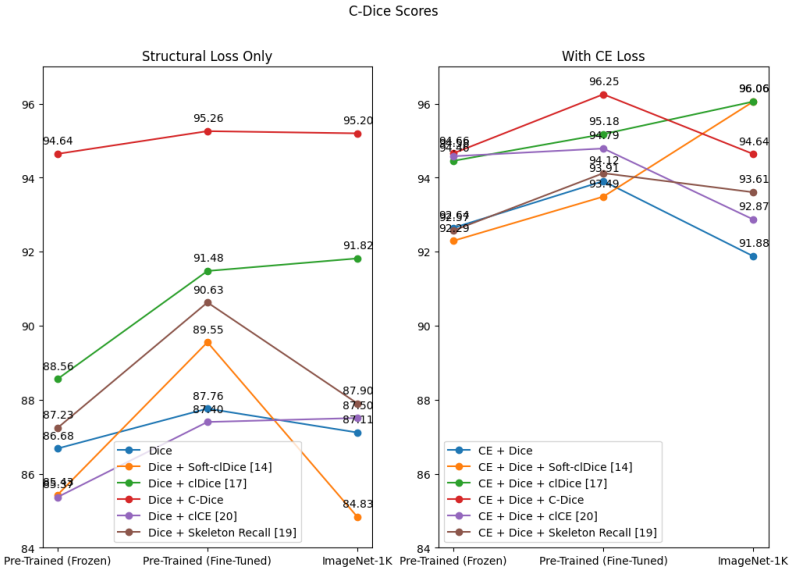


Fig. 12.3. C-Dice performance of proposed model

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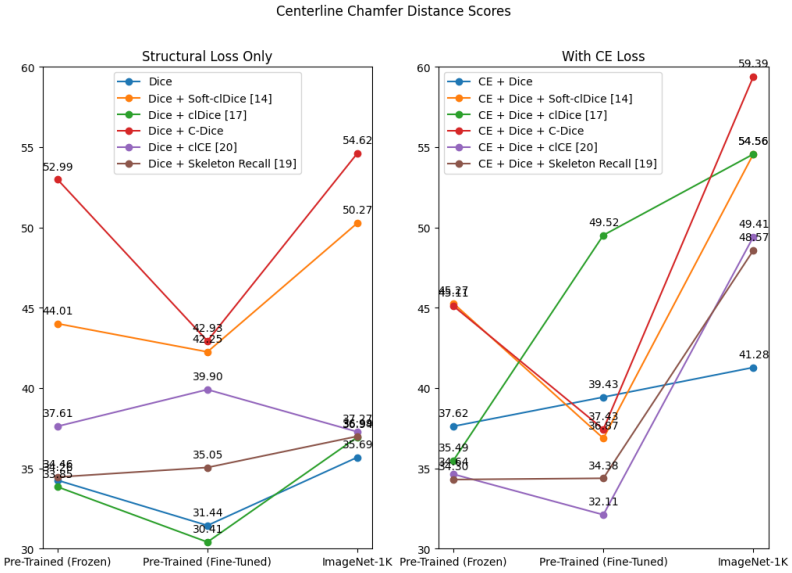


Fig. 12.4. Centerline Chamfer distance of proposed model

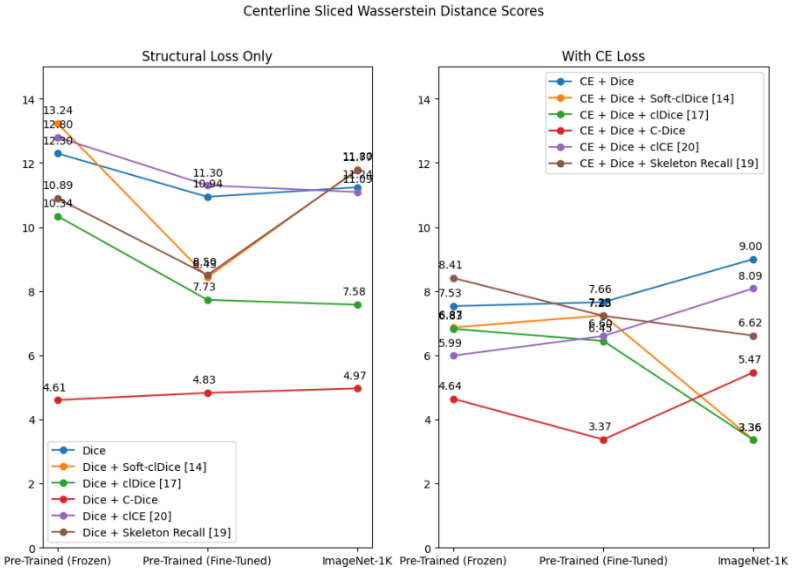


Fig. 12.5. Centerline sliced Wasserstein distance of proposed model

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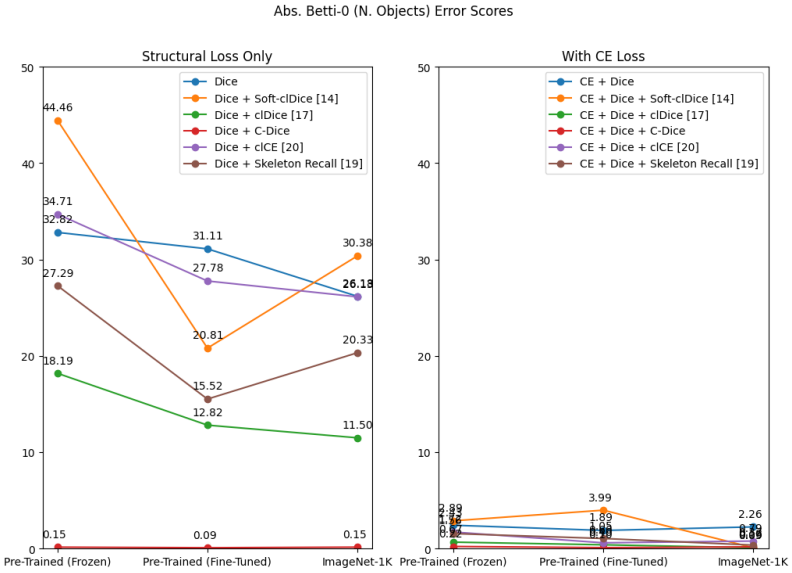


Fig. 12.6. Absolute Betti-0 error of proposed model

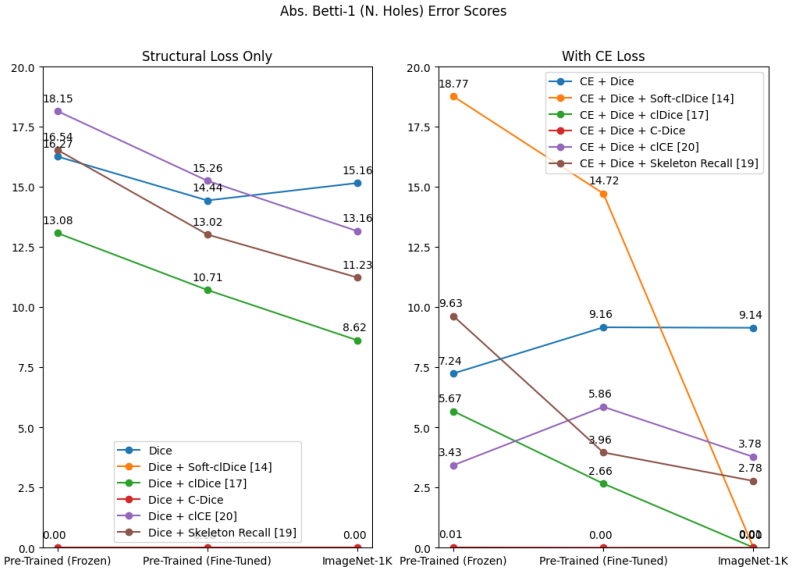


Fig. 12.7. Absolute Betti-1 error of proposed model

5.2.3 Multi-Task Models

The clDice score of the multi-task models was superior to most configurations of single-output model training, but the Dice score was slightly lower. C-Dice scores were comparable to single-output models trained with C-Dice loss or [17]'s clDice implementation. Although the SWD was comparable to single-output models, there was

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also a higher incidence of false positive detections of non-target structures than the best single-output models, as shown in the slightly higher β_0. Additionally, the clDice-trained multi-task models had markedly higher Chamfer distances. In conjunction with the low SWD values, this indicates high discrepancy in the total length of the centerline estimates. Existing multi-task models are therefore highly successful at placing the centerline prediction inside the ground truth area, but lag slightly in measures of centerline alignment. In terms of segmentation alignment, multi-task models surpass conventional Dice loss, but are surpassed by both clDice loss and our proposed C-Dice loss.

The non-clDice trained model achieved one of the lowest Chamfer distance scores overall, but performed worse than the other multi-task models on all other metrics, and was outperformed by the best single-output models on all metrics. This would indicate that using some version of clDice loss on EUS data is better than not.

Model	Dice Score	clDice Score	Vol. C-Dice Score	Chamfer Distance
Multi-UNet	**90.60%**	**98.00%**	**95.32%**	86.81
DTUNet [23]	89.97%	97.70%	94.76%	71.98
CL-Supervised UNet [27]	88.76%	96.17%	94.56%	**38.84**
clDice CasNet [24]	89.46%	94.18%	93.03%	72.85

Table 4.1. Segmentation performance and CD of multi-task models

Model	CL SWD	CL Abs. β_0 Error	CL Abs. β_1 Error
Multi-UNet	4.27	**0.03**	**0.00**
DTUNet [23]	**3.57**	0.05	**0.00**
CL-Supervised UNet [27]	5.47	0.14	**0.00**

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clDice CasNet [24]	4.45	0.04	**0.00**

Table 4.2. SWD and topological similarity of multi-task models' centerline estimates

Model	Vol. CL SWD	Vol. Abs. β_0 Error	Vol. Abs. β_1 Error
Multi-UNet	5.37	0.10	**0.00**
DTUNet [23]	4.42	0.08	**0.00**
CL-Supervised UNet [27]	5.90	0.10	**0.00**
clDice CasNet [24]	**4.41**	**0.02**	**0.00**

Table 4.3. Centerline SWD and topological similarity of multi-task models' segmentation volumes

5.3 External Datasets

For blood vessel segmentation in retinal fundus photographs, the Dice, clDice, and C-Dice scores were highly similar across loss functions, with total range of only 0.56%, 0.99%, and 0.94% respectively. The proposed C-Dice loss function did create slight improvement in Chamfer distance and topological connectivity, but the conventional Dice loss still had the highest Dice and C-Dice scores and lowest SWD. The results indicate a trade-off between segmenting narrower versus thicker blood vessels: ordinary Dice loss results in fewer false detections of very small blood vessels, but poorer connectivity of larger blood vessels. C-Dice loss produces more accurate segmentation of thicker blood vessels, but produces less complete segmentation of thin blood vessels. Therefore, in applications with predominantly thin segments, C-Dice is not always ideal.

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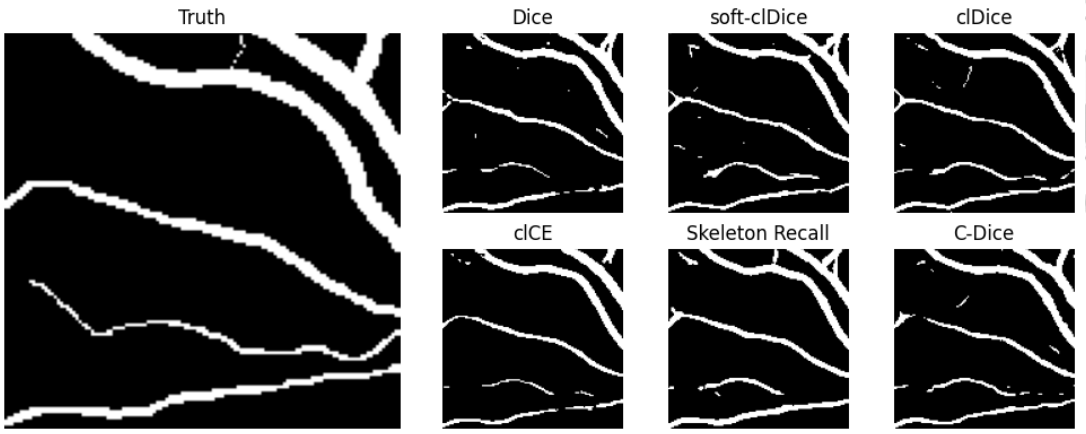


Fig. 13.1. Example segmentation of different loss functions for retinal fundus images

Loss Func	Dice Score	clDice Score	C-Dice Score	Chamfer Dist
A2	**75.88%**	79.30%	**80.20%**	6627.95
B2	75.32%	78.05%	79.32%	5347.44
C2	75.26%	78.30%	79.34%	4995.58
D2	74.48%	**79.61%**	80.03%	5044.71
E2	75.68%	79.03%	80.02%	5990.00
CE + C-Dice *(ours)*	75.35%	78.31%	79.26%	**4752.57**

Table 5.1. Performance of proposed model on retinal fundus images

Loss Func	SWD	Abs. β_0 Error	Abs. β_1 Error
A2	**7.12**	87.20	6.40
B2	8.27	90.60	6.25
C2	8.34	84.80	6.40
D2	7.25	94.0	8.5
E2	7.64	**79.2**	**5.75**
CE + C-Dice *(ours)*	7.61	83.20	5.80

Table 5.2. Performance of proposed model on retinal fundus images

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For heart chamber segmentation in echocardiography images, the proposed C-Dice loss slightly outperformed other loss functions in clDice score and SWD, with a Dice score only 0.01% behind conventional Dice loss. However, conventional Dice loss achieved a marginally larger C-Dice score than C-Dice loss. By examining the SWD, we see that C-Dice loss still produces slightly more aligned centerlines. The previous iterations of clDice, on the other hand, actually increased the SWD and still had a Dice score trade-off.

Loss Func	Dice Score	clDice Score	C-Dice Score	Chamfer Dist
A2	**96.51%**	98.32%	**96.44%**	158.29
B2	96.24%	96.96%	95.80%	**153.76**
C2	96.39%	98.27%	96.13%	177.75
D2	96.33%	97.11%	96.11%	154.73
E2	96.50%	98.32%	96.37%	161.71
CE + C-Dice *(ours)*	96.50%	**98.42%**	96.41%	161.08

Table 6.1. Performance of proposed model on echocardiography images

Loss Func	SWD	Abs. β_0 Error	Abs. β_1 Error
A2	5.14	0.01	0.01
B2	6.29	0.59	1.09
C2	5.57	0.01	**0.00**
D2	6.55	0.01	0.01
E2	5.07	**0.00**	**0.00**
CE + C-Dice *(ours)*	**4.96**	0.01	**0.00**

Table 6.2. Performance of proposed model on echocardiography images

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Chapter 6 Future Work

This work primarily investigates the modification of centerline loss methods to improve the segmentation of EUS landmark blood vessels. Although this work touches briefly on the performance of these adaptations in other segmentation tasks, only two imaging contexts and modalities are examined. Further work on C-Dice (and centerline loss functions as a whole) is necessary to properly characterize the relationship between segmentation radius and final training efficacy.

For EUS segmentation, future work should extend the size and diversity of the dataset in three main areas: First, this work concerns only a single landmark structure, the portal vein confluence, but there are other anatomic landmarks in the same region of interest. The biliary system (bile ducts and gallbladder) in particular is another landmark that is sometimes a target organ for EUS examinations. Second, this work was trained only on patients without major anatomic irregularities. Model performance should be verified on patients with more divergent anatomy, where navigational assistance may offer greater benefits to operators. Third, this work concerns only single-frame, one-shot detection. Expanding landmark detection to tracking across time could benefit segmentation quality in areas where landmarks fade into or out of frame, but more importantly, could assist the development of probe movement estimation and localization algorithms for EUS procedures.

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Chapter 7 Conclusion

The emerging application of deep learning in endoscopic ultrasound video shows potential to alleviate the mental load of performing EUS procedures and accelerating skill acquisition in EUS trainees, but training refinement for anatomic segmentation remains largely unexplored. We investigate the benefits to EUS landmark segmentation from centerline-based loss functions, proposing C-Dice loss, an expansion on existing methods. C-Dice improves the centerline alignment and topological connectivity across most model and loss configurations tested, including on two non-EUS datasets. We also quantify the benefits of backbone pre-adaptation to EUS data versus a popular pre-trained backbone, finding higher performance when using structural loss functions only, but near-equal performance when using cross-entropy loss. This represents an iteration towards improved EUS landmark segmentation and potential future EUS landmark tracking and probe localization, and our proposed loss function shows improvement in centerline alignment over existing methods.

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APPENDIX

1.1 Model Performance Tables

Dice Score	Backbone Configuration		
Loss	Pre-Adapted (Frozen)	Fine-Tuned	Not Pre-adapted
A1	86.99%	87.77%	86.99%
B1	83.37%	87.63%	84.49%
C1	87.70%	89.95%	90.13%
C-Dice *(ours)*	90.32%	90.82%	**91.17%**
A2	89.59%	90.55%	88.97%
B2	88.77%	90.02%	**92.34%**
C2	90.67%	90.89%	**92.34%**
CE + C-Dice *(ours)*	90.28%	91.88%	90.36%

Table 7.1. Dice performance of proposed model

clDice Score	Backbone Configuration		
Loss	Pre-Adapted (Frozen)	Fine-Tuned	Not Pre-adapted
A1	75.72%	76.76%	76.84%
B1	83.96%	88.30%	81.82%
C1	79.49%	83.77%	84.55%
C-Dice *(ours)*	94.66%	**95.92%**	95.30%
A2	87.45%	89.72%	86.57%
B2	91.80%	91.85%	**97.06%**
C2	91.54%	92.89%	**97.06%**
CE + C-Dice *(ours)*	94.89%	96.87%	94.06%

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Table 7.2. clDice performance of proposed model

C-Dice Score	Backbone Configuration		
Loss	Pre-Adapted (Frozen)	Fine-Tuned	Not Pre-adapted
A1	86.68%	87.75%	87.11%
B1	85.43%	89.55%	84.83%
C1	88.56%	91.48%	91.82%
C-Dice *(ours)*	94.64%	**95.26%**	95.19%
A2	92.64%	93.91%	91.88%
B2	92.29%	93.49%	96.06%
C2	94.46%	95.18%	96.06%
CE + C-Dice *(ours)*	94.66%	**96.25%**	94.64%

Table 7.3. C-Dice performance of proposed model

CD	Backbone Configuration		
Loss	Pre-Adapted (Frozen)	Fine-Tuned	Not Pre-adapted
A1	1358.47	795.36	35.64
B1	1621.48	42.25	50.27
C1	33.85	**30.41**	36.94
C-Dice *(ours)*	52.99	42.93	54.62
A2	234.71	302.32	41.29
B2	45.67	36.87	54.56
C2	**35.49**	49.52	54.56
CE + C-Dice *(ours)*	45.11	37.43	59.38

Table 7.4. Centerline Chamfer distance performance of proposed model

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SWD	Backbone Configuration		
Loss	Pre-Adapted (Frozen)	Fine-Tuned	Not Pre-adapted
A1	12.29	10.94	11.24
B1	13.24	8.43	11.80
C1	10.34	7.73	7.58
C-Dice *(ours)*	**4.61**	4.83	4.96
A2	7.54	7.65	8.96
B2	6.87	7.25	**3.36**
C2	6.83	6.45	**3.36**
CE + C-Dice *(ours)*	4.64	3.37	5.47

Table 7.5. Centerline sliced Wasserstein distance of proposed model

Abs. β_0 Error	Backbone Configuration		
Loss	Pre-Adapted (Frozen)	Fine-Tuned	Not Pre-adapted
A1	32.77	31.10	26.36
B1	44.46	20.81	30.38
C1	18.19	13.32	12.56
C-Dice *(ours)*	**0.00**	**0.00**	0.01
A2	2.65	1.96	2.48
B2	3.26	4.41	**0.02**
C2	0.63	0.32	**0.02**
CE + C-Dice *(ours)*	0.11	0.03	0.06

Table 7.6. Absolute Betti-0 error of proposed model

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Abs. β_1 Error	Backbone Configuration		
Loss	Pre-Adapted (Frozen)	Fine-Tuned	Not Pre-adapted
A1	17.95	14.62	16.03
B1	64.44	55.74	54.80
C1	13.73	11.46	9.21
C-Dice *(ours)*	**0.00**	**0.00**	**0.00**
A2	8.19	10.34	11.18
B2	18.95	15.94	**0.00**
C2	6.20	3.04	**0.00**
CE + C-Dice *(ours)*	**0.00**	**0.00**	**0.00**

Table 7.7. Absolute Betti-1 error of proposed model

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